MA3521 — A Forced 3×3 IVP

Name:

Solve the IVP and determine behavior as $t \to \infty$.

$$\mathbf{y}' = A\mathbf{y} + e^{-2t}\begin{pmatrix}-4\\-4\\-4\end{pmatrix} + t\begin{pmatrix}2\\2\\0\end{pmatrix} \qquad A = \begin{pmatrix}-4 & 2 & 4\\-2 & 1 & 3\\-4 & 3 & 3\end{pmatrix} \qquad \mathbf{y}(0) = \mathbf{y}_0 = \begin{pmatrix}3\\1\\4\end{pmatrix}$$

a. Eigenvalues and eigenvectors

$\lambda_1 =$ $\lambda_2 =$ $\lambda_3 =$

$V = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3] =$

b. Real complementary solution $\mathbf{y}_c(t) = c_1\mathbf{y}_1(t) + c_2\mathbf{y}_2(t) + c_3\mathbf{y}_3(t)$

Write down three linearly independent real solutions

$\mathbf{y}_1(t) =$ $\mathbf{y}_2(t) =$ $\mathbf{y}_3(t) =$

c. Exponential forcing $\mathbf{f}_1(t) = e^{-2t}(-4, -4, -4)^T$

Try $\mathbf{y}_{p_1}(t) = \mathbf{a}e^{-2t}$ $\mathbf{y}'_{p_1}(t) =$

Substitute into $\mathbf{y}'_{p_1} = A\mathbf{y}_{p_1} + \mathbf{f}_1$ and cancel the exponential.

$\mathbf{a} =$ $\mathbf{y}_{p_1}(t) =$

d. Linear forcing $\mathbf{f}_2(t) = t(2,2,0)^T$

Try $\mathbf{y}_{p_2}(t) = \mathbf{b}t + \mathbf{d}$ $\mathbf{y}'_{p_2}(t) =$

Substitute into $\mathbf{y}'_{p_2} = A\mathbf{y}_{p_2} + \mathbf{f}_2$ and match the coefficient of t and the constant term and solve

$\mathbf{y}_{p_2}(t) =$

e. Particular and general solutions

$\mathbf{y}_p(t) = \mathbf{y}_{p_1}(t) + \mathbf{y}_{p_2}(t) =$

$\mathbf{y}(t) = \mathbf{y}_c(t) + \mathbf{y}_p(t) =$

f. Initial condition in GS $\mathbf{y}(t) = \mathbf{y}_c(t) + \mathbf{y}_{p_1}(t) + \mathbf{y}_{p_2}(t)$

Write down the augmented matrix you would row reduce to solve for the constants $\mathbf{c}$ to satisfy $\mathbf{y}(0) = xxx$.

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g. IVP solution and leading behavior

Solve for $\mathbf{c}$ and fill in the asymptotic behavior.